t2g square lattice tight-binding derivation

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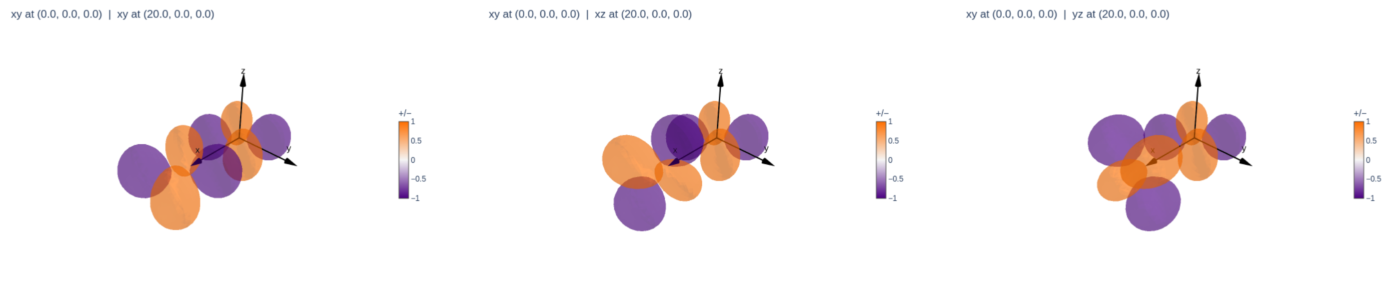
1 Single Layer

We look at hopping from all orbitals along the two x-y directions.

1.1 xy band

1.1.1 x-direction



(a) d_{xy}–d_{xy} (b) d_{xy}–d_{xz} (c) d_{xy}–d_{yz}

Figure 1: t2g orbital overlaps for nearest-neighbour hopping along x.

We look at the slater-koster integrals. The slater direction cosines are

$$l = \frac{\boldsymbol{R} \cdot \hat{x}}{|\boldsymbol{R}|} \quad m = \frac{\boldsymbol{R} \cdot \hat{y}}{|\boldsymbol{R}|} \quad n = \frac{\boldsymbol{R} \cdot \hat{z}}{|\boldsymbol{R}|} \tag{1}$$

$$\tag{2}$$

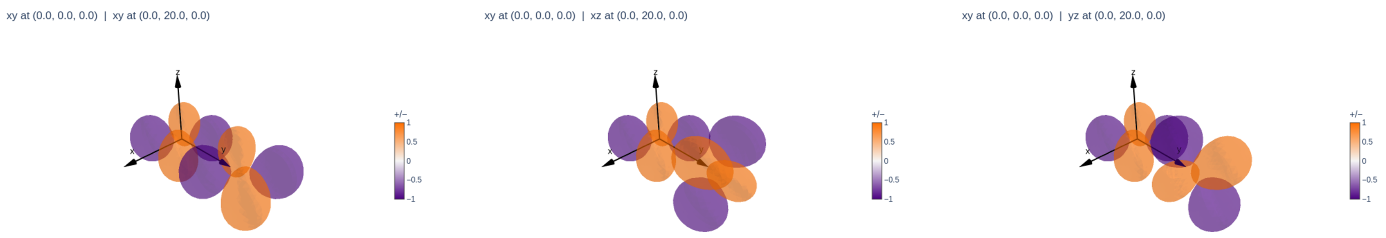
In the x direction this is just.

$$l = 1 \quad m = 0 \quad n = 0 \tag{3}$$

This simplifies the xy-xy, xy-xz, and xy-yz integrals to

$$t_{xy,xy} = V_{dd\pi} \quad t_{xy,yz} = 0 \quad t_{xy,xz} = 0 \tag{4}$$

1.1.2 y-direction



(a) d_{xy}–d_{xy} (b) d_{xy}–d_{xz} (c) d_{xy}–d_{yz}

Figure 2: t2g orbital overlaps for nearest-neighbour hopping along y.

For the y-direction the slater direction cosines are

$$l = 0 \quad m = 1 \quad n = 0 \tag{5}$$

and the integrals are

$$t_{xy,xy} = V_{dd\pi} \quad t_{xy,yz} = 0 \quad t_{xy,xz} = 0 \tag{6}$$

1.1.3 Tight Binding Hamiltonian

Putting this all together we get this hopping term in the hamiltonian and apply the Fourier transformation $c_i = \frac{1}{\sqrt{N}} \sum_k e^{i\boldsymbol{k}\cdot\boldsymbol{R}_i} c_{\boldsymbol{k}}$

$$\begin{aligned}
H &= t \sum_{R,\delta} c^\dagger_{i+\delta} c_i & (7)\\
&= \frac{t}{N} \sum_{R,\delta} \sum_{q,k} e^{-i\boldsymbol{q}\cdot(\boldsymbol{R}_i+\boldsymbol{\delta})} e^{i\boldsymbol{k}\cdot\boldsymbol{R}_i} c^\dagger_{\boldsymbol{q}} c_{\boldsymbol{k}}\\
&= \frac{t}{N} \sum_{\delta} \sum_{q,k} \delta_{qk} e^{-i\boldsymbol{q}\cdot\boldsymbol{\delta}} c^\dagger_q c_k\\
&= t \sum_{\delta} \sum_{k} e^{-iq\cdot\delta} c^\dagger_k c_k\\
&= t \sum_{k} \left[e^{-ik_x a} + e^{-ik_y a} + e^{ik_x a} + e^{ik_y a} \right] c^\dagger_k c_k\\
&= 2V_{dd\pi} \sum_{k} \left[\cos(k_x a) + cos(k_y a)\right] c^\dagger_k c_k
\end{aligned}$$

1.1.4 Next Nearest Neighbor Hopping

Now lets consider this one for the xy band. The hopping vector is now $(a\hat{x} + a\hat{y})$. Our directional cosines are

$$l = 1/\sqrt{2} \quad m = 1/\sqrt{2} \quad n = 0 \tag{8}$$

and the non zero integrals are

$$t_{xy-xy} = \frac{3}{4} V_{dd\sigma} + \frac{1}{4} V_{dd\delta} \tag{9}$$

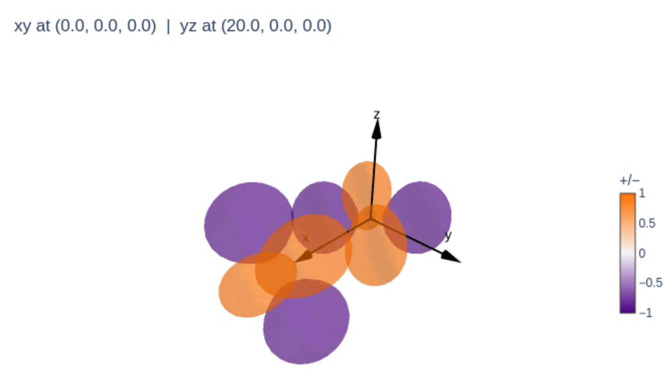
it turns out this is the result for all directions i.e $(\pm a\hat{x} + \pm a\hat{y})$ our tight binding hamiltonian can then be written as.

$$\begin{aligned}
H &= t_{nnn} \sum_{\delta} \sum_{k} e^{-ik\cdot\delta} c_k^\dagger c_k \\
&= t_{nnn} \sum_{k} \left[e^{-ia(k_x+k_y)} + e^{-ia(k_x-k_y)} + e^{-ia(-k_x+k_y)} + e^{-ia(-k_x-k_y)} \right] c_k^\dagger c_k \\
&= t_{nnn} \sum_{k} \left[2\cos(k_x a + k_y a) + 2\cos(k_x a - k_y a) \right] c_k^\dagger c_k \\
&= 4t_{nnn} \sum_{k} \cos(k_x a)\cos(k_y a) c_k^\dagger c_k
\end{aligned} \tag{10}$$

Using the sum to product $\cos(A+B) + \cos(A-B) = 2\cos A \cos B$

1.2 yz band

To look at the yz band we apply the following permutation. Its helpful to look at this diagram



From xy we get yz by permuting the axis like $x \to y \to z \to x$, which means our direction cosines become $l \to m \to n \to l$

1.2.1 x direction

$$l = 0 \quad m = 1 \quad n = 0 \tag{11}$$

$$t_{yz,yz} = V_{dd\pi} \quad t_{yz,xy} = 0 \quad t_{yz,xz} = 0 \tag{12}$$

The hamiltonian is

$$\begin{aligned}
H &= t \sum_{\delta} \sum_{k} e^{-ik\cdot\delta} c_k^\dagger c_k \\
&= t \sum_{k} \left[e^{-ik_x a} + e^{ik_x a} \right] c_k^\dagger c_k \\
&= 2V_{dd\pi} \sum_{k} \cos(k_x a) c_k^\dagger c_k
\end{aligned} \tag{13}$$

1.2.2 y direction

$$l = 0 \quad m = 0 \quad n = 1 \tag{14}$$

$$t_{yz,yz} = V_{dd\delta} \quad t_{yz,xy} = 0 \quad t_{yz,xz} = 0 \tag{15}$$

The derivation for the energy is the same as above just in y direction.

1.3 Energy for band

Hence the total energy is

$$\varepsilon_{yz} = -2t_1 \cos(k_x a) - 2t_\delta \cos(k_y a) \tag{16}$$

1.4 xz band

The same formalism applies.

1.5 x dir

$$l = 0 \quad m = 0 \quad n = 1 \tag{17}$$

which gives the nonzero integral $V_{dd\delta}$

1.6 y dir

$$l = 1 \quad m = 0 \quad n = 0 \tag{18}$$

which gives the nonzero integral $V_{dd\pi}$.

1.7 Energy For Band

$$\varepsilon_{xz} = -2t_\delta \cos(k_x a) - 2t_1 \cos(k_y a) \tag{19}$$